How Does AI Watermarking Work?

# A Close Look at SynthID

**Emanuele Fabbiani** 🦫

Codemotion Milan · 28–29 October 2026

# Detecting AI text is hard.

## Jan 2023

**OpenAI releases a detector for AI-written text**

## Jul 2023

**Shut down for its low accuracy**

OpenAI, New AI classifier for indicating AI-written text, 2023







Anthropic, How Claude's text watermark works, 2026; OpenAI, Our approach to EU text provenance rules, 2026

# How did we get there?

01 External detectors

02 Special characters

03 Red and green lists

04 Soft lists

05 **Tournament sampling** SYNTHID

CHAPTER 1 OF 6

01

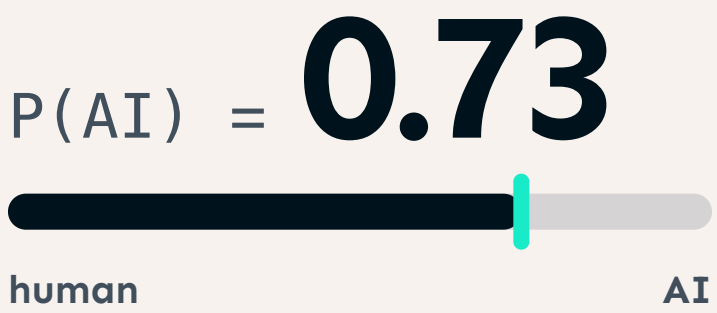
# External detectors

## Any finished text

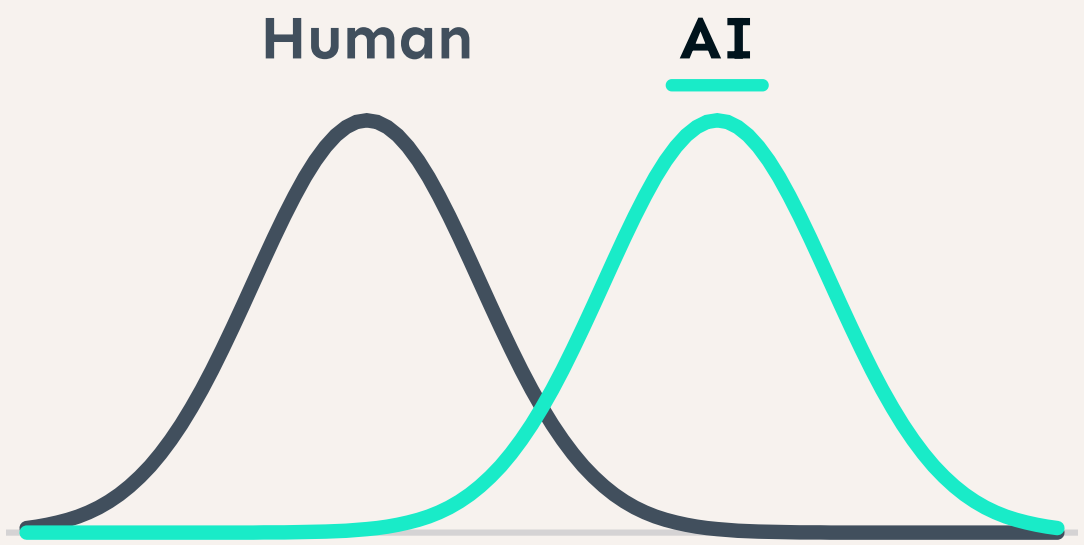




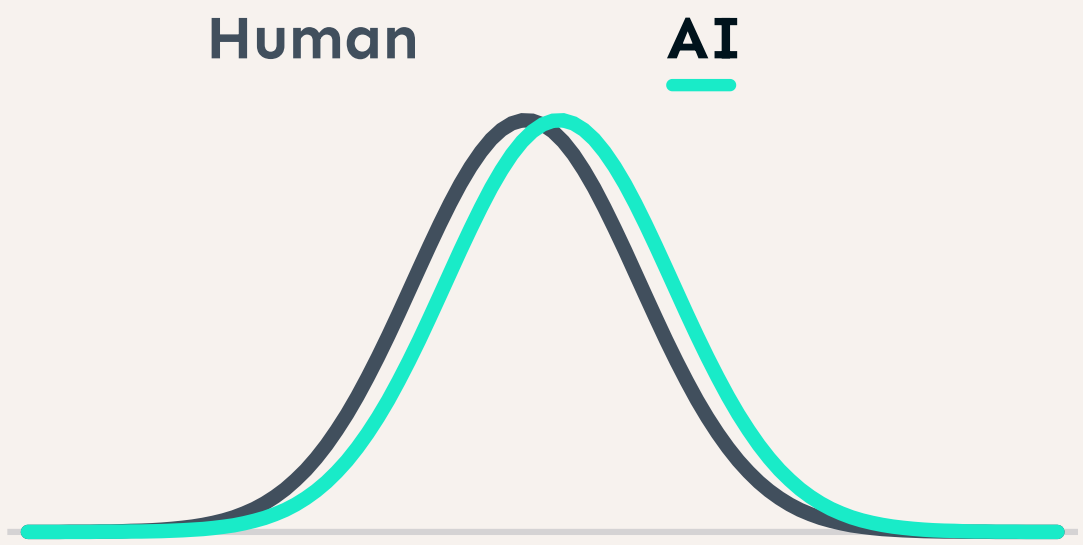
Classifier or perplexity test



WHEN MODELS WROTE BADLY




TODAY




Sadasivan et al., Can AI-Generated Text be Reliably Detected?, 2025

1

# Add the signal at generation time.

CHAPTER 2 OF 6

02

# Special characters

WHAT THE READER SEES

Watermarks hide in plain sight

WHAT THE BYTES SAY

Watermarks U+200B hide U+2009 in U+200B plain U+2009 sight

Sato et al., Embarrassingly Simple Text Watermarks, 2023

```
import re, unicodedata

text = unicodedata.normalize("NFKC", text)
text = re.sub(r"[\u200b-\u200f\u2060\ufeff]", "", text)
```

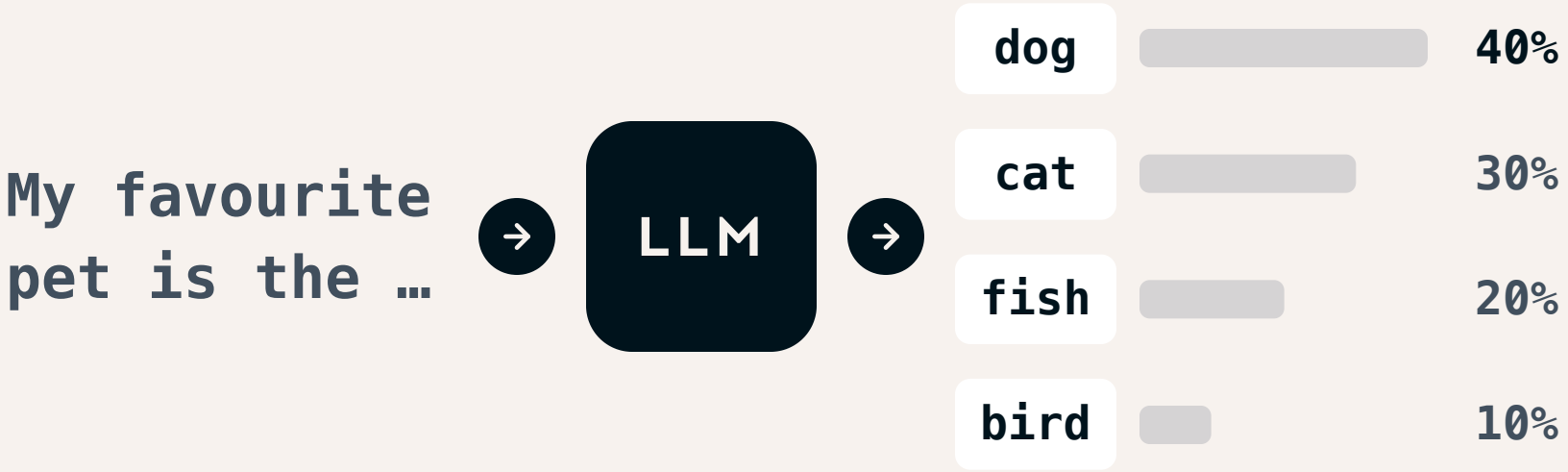
2

# Put the signal in the word choice.

CHAPTER 3 OF 6

03

# Red and green lists


My favourite pet is the …
LLM
dog
40%
cat
30%
fish
20%
bird
10%


My favourite pet is the …
LLM
dog 40%
cat 30%
fish 20%
bird 10%
RANDOM DRAW
… the dog

My favourite pet is **the** ?

## 1 Seed

**the** last token

+

secret

**hash(last token, secret)**

seed = 4821

Kirchenbauer et al., A Watermark for Large Language Models, 2023

# My favourite pet is the ?

## 1 Seed

the last token

+

secret

**hash(last token, secret)**

seed = 4821

## 2 Split the vocabulary




Kirchenbauer et al., A Watermark for Large Language Models, 2023

# My favourite pet is the ?

## 1 Seed

the last token

+

secret

**hash(last token, secret)**

seed = 4821

## 2 Split the vocabulary

| | | |
|---|---|---|
| cat | bird | car |
| song | book | plan |
| dog | fish | idea |
| river | city | word |

## 3 Sample only green

| | | |
|---|---|---|
| ~~dog~~ | | ~~40%~~ |
| cat | | 30% |
| ~~fish~~ | | ~~20%~~ |
| bird | | 10% |

# My favourite pet is the cat

Kirchenbauer et al., A Watermark for Large Language Models, 2023

## same key, recolour every token

HUMAN TEXT

dogs are loyal and love long walks daily

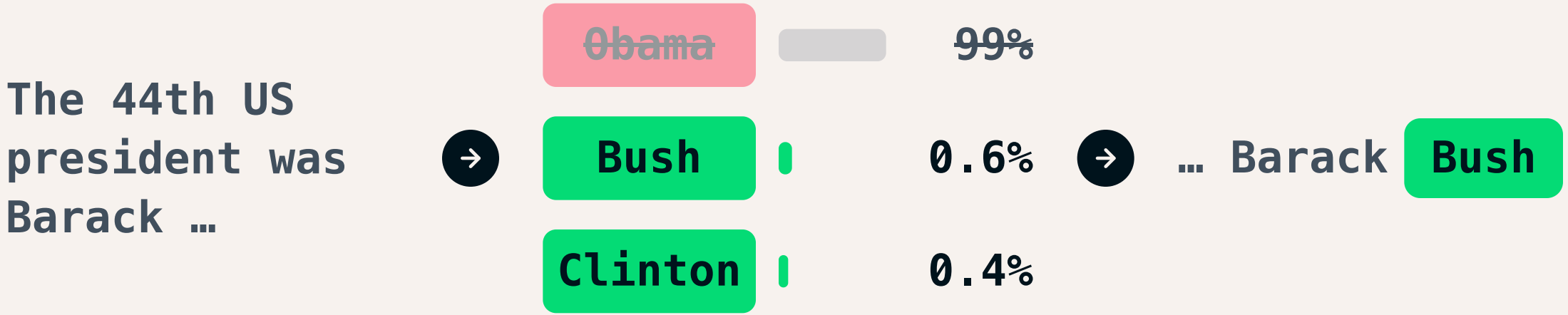
4/8
green tokens

WATERMARKED TEXT

cats are calm and purr on my lap

7/8
green tokens

Kirchenbauer et al., A Watermark for Large Language Models, 2023




Kirchenbauer et al., A Watermark for Large Language Models, 2023

3

# Step aside when the model is sure (low entropy).

CHAPTER 4 OF 6

04

# Soft lists

| | ODDS | | PROBABILITY |
|---|---|---|---|
| dog | 4 | → | 4/10 = 40% |
| cat | 3 | → | 3/10 = 30% |
| fish | 2 | → | 2/10 = 20% |
| bird | 1 | → | 1/10 = 10% |

probability = odds ÷ sum of all odds

My favourite pet is the ?

## 1 Boost the green odds

green odds × δ = 7

red odds × 1

## 2 New probabilities

| | BEFORE | | AFTER |
|---|---|---|---|
| dog | 40% | → | 12% |
| cat | 30% | → | 62% |
| fish | 20% | → | 6% |
| bird | 10% | → | 21% |

My favourite pet is the cat

Kirchenbauer et al., A Watermark for Large Language Models, 2023

The 44th US president was Barack ?

**1** Boost the green odds

green odds × δ = 7

red odds × 1

**2** New probabilities

| | BEFORE | | AFTER |
|---|---|---|---|
| Obama | 99% | → | 93% |
| Bush | 0.6% | → | 4% |
| Clinton | 0.4% | → | 3% |

The 44th US president was Barack Obama

Kirchenbauer et al., A Watermark for Large Language Models, 2023

# ~800

tokens are enough to detect AI content
after strong human paraphrasing

Kirchenbauer et al., On the Reliability of Watermarks for Large Language Models, 2024

4

# To preserve quality, don't change the LLM's output distribution.

CHAPTER 5 OF 6

05

# Tournament sampling

My favourite pet is the ?

## 1 Sample the candidates




| | | | |
|---|---|---|---|
| dog | cat | dog | fish |
| cat | dog | bird | cat |

$2^3$ = 8 candidates, likely words more often

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

My **favourite pet is the** ?

## 2 Seed, once per round

```
hash(last 4 tokens, key, round=1) = 4821
hash(last 4 tokens, key, round=2) = 1907
hash(last 4 tokens, key, round=3) = 6350
```

## 3 Score every candidate

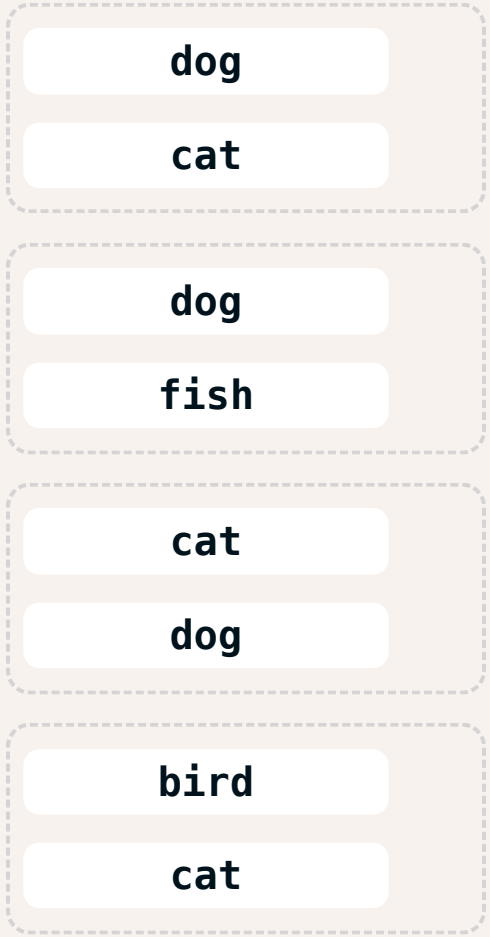
| | R1 | R2 | R3 |
|---|---|---|---|
| dog | 0 | 1 | 0 |
| cat | 1 | 0 | 1 |
| fish | 1 | 1 | 0 |
| bird | 0 | 1 | 1 |

SynthID uses 30 rounds

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

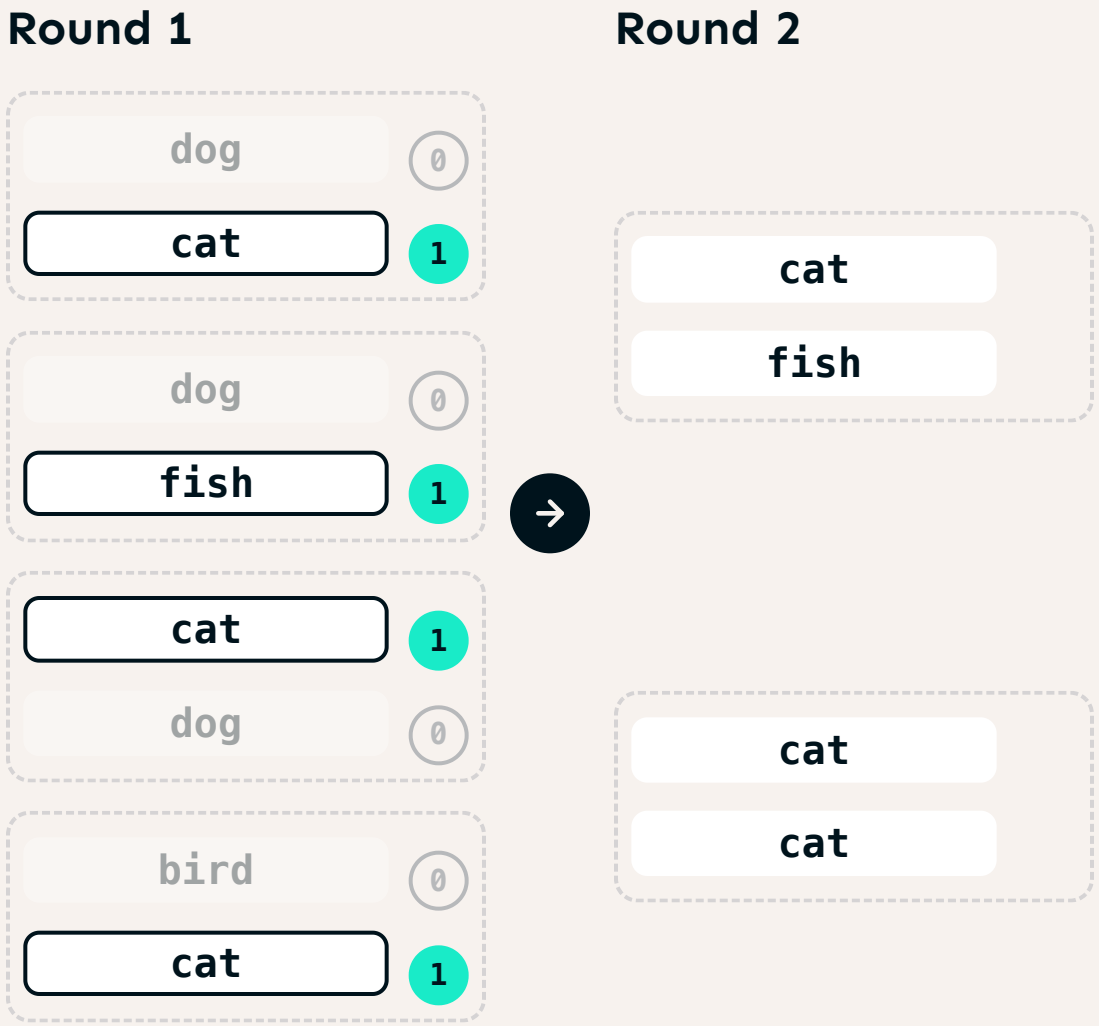
# 4 Tournament sampling

## Round 1



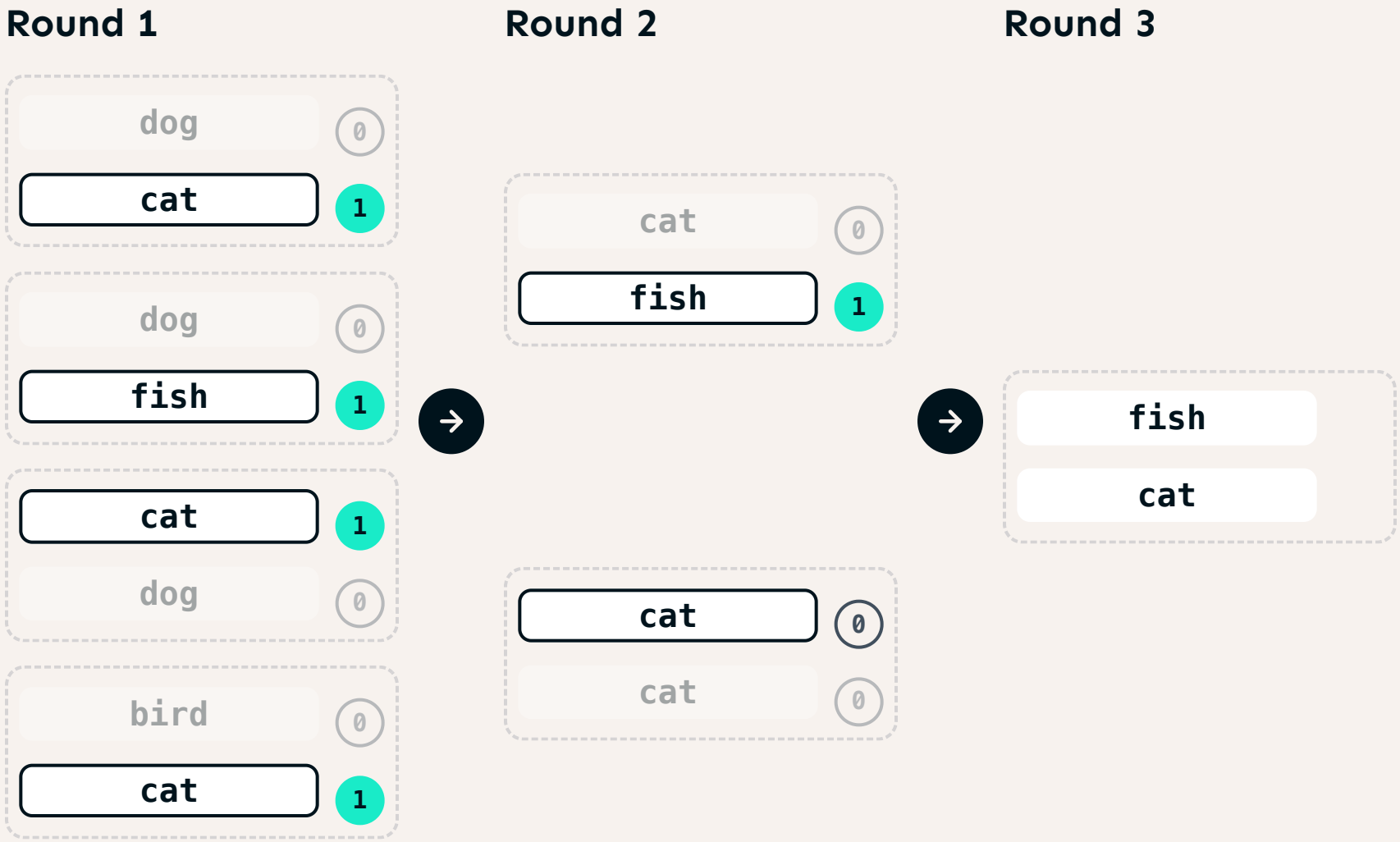

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

# 4 Tournament sampling



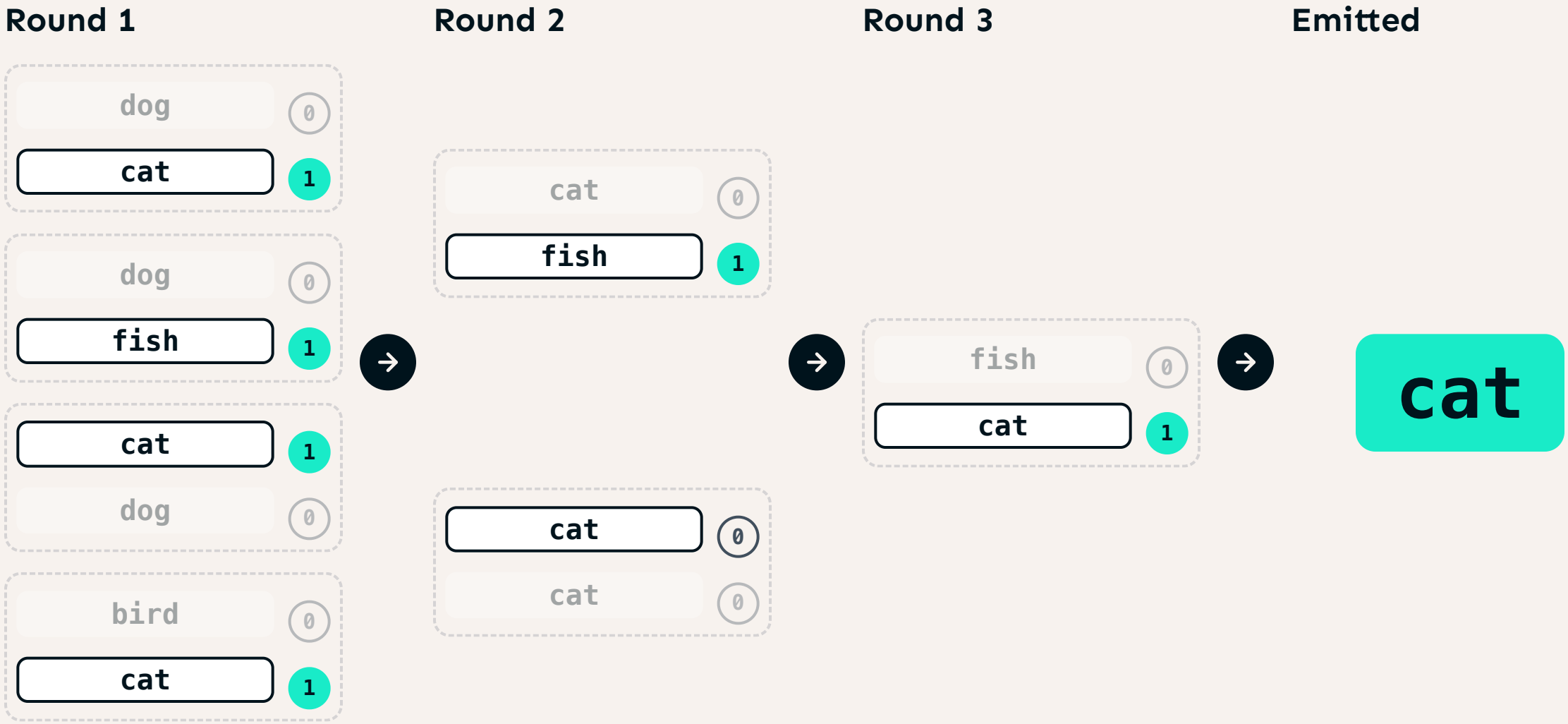

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

# 4 Tournament sampling




Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

# 4 Tournament sampling




Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024




Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

My favourite pet is the cat, and your favourite pet is the …

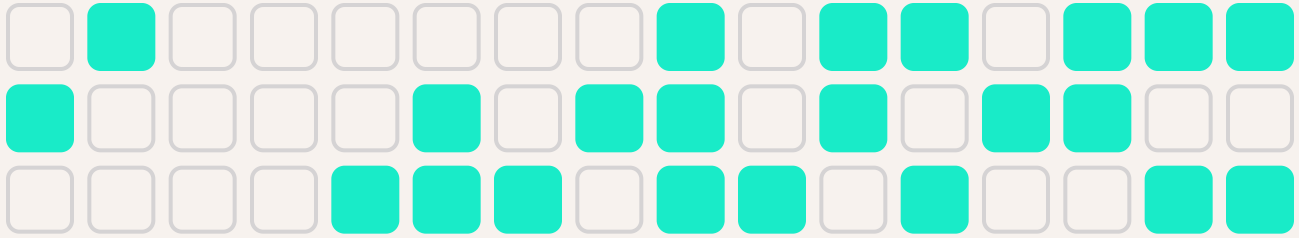
**1st time** `favourite pet is the` → 4821 · 1907 · 6350 seeds → tournament → **cat**

**2nd time** `favourite pet is the` → 4821 · 1907 · 6350 same seeds, same tilt → sample normally → **dog**

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

## same key, recompute every score

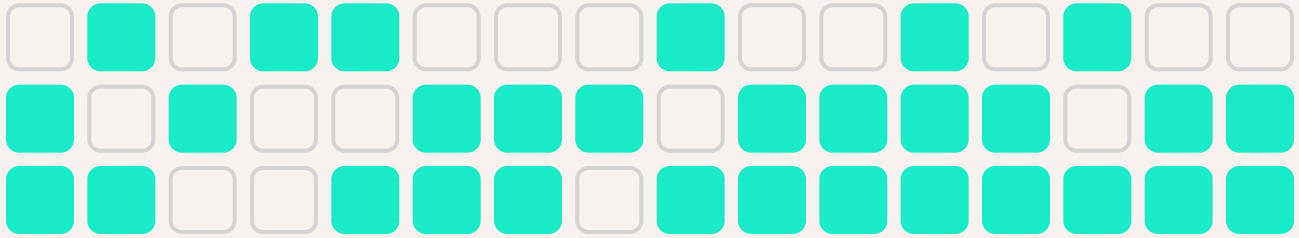
HUMAN TEXT



**0.46**

mean score

WATERMARKED TEXT



**0.63**

mean score

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

**+0.57%**

latency per token

15.527 → 15.615 ms on Gemma 7B

**0.01%**

thumbs-up gap

over ~20M Gemini responses

**95%**

true-positive rate

at 1% false positives, abstaining on uncertain texts

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

CHAPTER 6 OF 6

06

# Limits

FREE CHOICE CARRIES SIGNAL · FORCED BY SYNTAX DOESN'T

TOKENS NEEDED TO DETECT

## Prose

my cat loves long naps in the sun

few

## Code

```
for x in range ( 10 ) :
```

many

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

# A full rewrite or a translation erases it.

OpenAI, Understanding the source of what we see and hear online, 2024

# Only the key holder can check for it.

Google, SynthID Detector — a new portal to help identify AI-generated content, 2025

| LESSON | ATTEMPT | WHAT IT TAUGHT US |
|---|---|---|
| 1 | External detectors | **Add the signal at generation time** |
| 2 | Special characters | **Put the signal in the word choice** |
| 3 | Red and green lists | **Step aside when the model is sure** |
| 4 | Soft lists | **Don't change the output distribution** |

**Tournament sampling** SYNTHID

ABOUT ME

# Emanuele Fabbiani

## Engineering Manager at TeamSystem



**Founder** at xtream and Reaidy, both acquired by TeamSystem in 2025.

**Professor** in AI at the Catholic University of Milan; guest lecturer at PoliMI, EPFL and Politechnika Wrocławska.

**PhD** in Applied AI, with 9 peer-reviewed papers and 180+ citations.

**Speaker** at 40+ international conferences, including AMLD Lausanne, ODSC London, WeAreDevelopers Berlin, PyData Berlin, Paris and Milan, PyCon Italy, Swiss Python Summit, Codemotion.

**Connect on LinkedIn**

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**[28]** nohypeai, Watermarking Playground, 2026